

TARGETSQL_BIBHU_PRASAD_PANI

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1.A. Data type of all columns in the "customers" table.

```
SELECT column_name, data_type
FROM `Target_Dataset.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers';
```

column_name ▼	data_type ▼
customer_id	STRING
customer_unique_id	STRING
customer_zip_code_prefix	INT64
customer_city	STRING
customer_state	STRING

1.B. Get the time range between which the orders were placed.

```
SELECT MIN(order_purchase_timestamp) AS first_order,
       MAX(order_purchase_timestamp) AS last_order
FROM Target_Dataset.orders;
```

first_order ▼	last_order ▼
2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

1.C. Count the Cities & States of customers who ordered during the given period.

```
SELECT count(DISTINCT c.customer_city) AS city_cnt,
       count(DISTINCT c.customer_state) AS state_cnt
FROM Target_Dataset.customers AS c
INNER JOIN Target_Dataset.orders AS o
ON c.customer_id = o.customer_id;
```

city_cnt ▼	state_cnt ▼
4119	27

2. In-depth Exploration:

2.A. Is there a growing trend in the no. of orders placed over the past years?

```
WITH trend AS (SELECT Year, no_of_orders,
                      LAG(no_of_orders) OVER(ORDER BY Year) AS lagg
                FROM(
                      SELECT EXTRACT(YEAR FROM order_purchase_timestamp) AS Year,
                             COUNT(order_id) AS no_of_orders
                      FROM Target_Dataset.orders
                      GROUP BY Year
                      ORDER BY Year) AS temp_table
                ORDER BY Year)
SELECT Year,
       ROUND(ifnull((no_of_orders-lagg)/lagg,0)*100,2) AS percentage_change
FROM trend;
```

Year ▼	percentage_change
2016	0.0
2017	13608.51
2018	19.76

2.B. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
WITH trend AS (SELECT Month, no_of_orders,
                      LAG(no_of_orders) OVER(ORDER BY Month) AS lagg
                FROM(
                      SELECT EXTRACT(Month FROM order_purchase_timestamp) AS Month,
                             COUNT(order_id) AS no_of_orders
                      FROM Target_Dataset.orders
                      GROUP BY Month
                      ORDER BY Month) AS temp_table
                ORDER BY Month)
SELECT Month,
       ROUND(ifnull((no_of_orders-lagg)/lagg,0)*100,2) AS percentage_change
```

FROM trend;

Month ▼	percentage_change
1	0.0
2	5.44
3	16.28
4	-5.56
5	13.16
6	-10.98
7	9.63
8	5.09
9	-60.3
10	15.19
11	52.13
12	-24.79

2.C. During what time of the day, do the Brazilian customers mostly place their orders?
(Dawn, Morning, Afternoon or Night).

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

```
WITH Final AS (SELECT order_purchase_timestamp,order_id,
CASE
WHEN EXTRACT(Hour FROM order_purchase_timestamp) <= 6
THEN "Dawn"
WHEN EXTRACT(Hour FROM order_purchase_timestamp) <= 12
THEN "Mornings"
WHEN EXTRACT(Hour FROM order_purchase_timestamp) <= 18
THEN "Afternoon"
WHEN EXTRACT(Hour FROM order_purchase_timestamp) <= 23
THEN "Night"
ELSE "Unknown"
END AS time_of_the_day
FROM Target_Dataset.orders)

SELECT time_of_the_day,count(order_id) as no_of_orders
FROM Final
```

```
GROUP BY time_of_the_day
ORDER BY no_of_orders DESC
LIMIT 1;
```

time_of_the_day ▼	no_of_orders ▼
Afternoon	38135

3. Evolution of E-commerce orders in the Brazil region.

3.A. Get the month on month no. of orders placed in each state.

```
SELECT c.customer_state,
       EXTRACT(MONTH FROM o.order_purchase_timestamp) as Month, count(o.order_id) as
no_of_orders
FROM Target_Dataset.orders AS O
INNER JOIN Target_Dataset.customers as C
ON O.customer_id = C.customer_id
GROUP BY C.customer_state, Month
ORDER BY C.customer_state, Month;
```

Row	customer_state ▼	Month ▼	no_of_orders ▼
1	AC	1	8
2	AC	2	6
3	AC	3	4
4	AC	4	9
5	AC	5	10
6	AC	6	7
7	AC	7	9
8	AC	8	7
9	AC	9	5
10	AC	10	6

3.B. How are the customers distributed across all the states?

```
SELECT customer_state, count(DISTINCT customer_id) as no_of_customers
FROM Target_Dataset.customers
GROUP BY customer_state
```

ORDER BY no_of_customers DESC;

Row	customer_state	no_of_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020

4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

4.A. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

```
WITH tempt1 as
    (SELECT EXTRACT(YEAR FROM o.order_purchase_timestamp) as
Year,sum(p.payment_value) as cost
FROM Target_Dataset.orders AS o
INNER JOIN Target_Dataset.payments AS p
ON o.order_id = p.order_id
WHERE EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8 AND
EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017,2018)
GROUP BY Year
ORDER BY Year),
tempt2 AS
    (SELECT *, LAG(cost) OVER(ORDER BY Year) as lagg
FROM tempt1)
SELECT Year,ROUND(ifnull(((cost-lagg)/lagg)*100,0),2) as Percentage_increase
FROM tempt2
ORDER BY Year;
```

Year ▼	Percentage_increase ▼
2017	0.0
2018	136.98

4.B. Calculate the Total & Average value of order price for each state.

```

SELECT c.customer_state,
       ROUND(sum(oi.price),2) as total_price,
       ROUND(sum(oi.price)/count(DISTINCT oi.order_id),2) as avg_price
FROM Target_Dataset.order_items as oi
INNER JOIN Target_Dataset.orders AS o
ON oi.order_id = o.order_id
INNER JOIN Target_Dataset.customers as c
ON c.customer_id = o.customer_id
GROUP BY c.customer_state;

```

Row	customer_state ▼	total_price ▼	avg_price ▼
1	SP	5202955.05	125.75
2	RJ	1824092.67	142.93
3	PR	683083.76	136.67
4	SC	520553.34	144.12
5	DF	302603.94	142.4
6	MG	1585308.03	137.33
7	PA	178947.81	184.48
8	BA	511349.99	152.28
9	GO	294591.95	146.78
10	RS	750304.02	138.13

4.C. Calculate the Total & Average value of order freight for each state.

```

SELECT c.customer_state,ROUND(sum(oi.freight_value),2) as
total_freight_value,ROUND(sum(oi.freight_value)/count(DISTINCT oi.order_id),2) as avg_freight_value
FROM Target_Dataset.order_items as oi
INNER JOIN Target_Dataset.orders AS o
ON oi.order_id = o.order_id
INNER JOIN Target_Dataset.customers as c
ON c.customer_id = o.customer_id
GROUP BY c.customer_state;

```

Row	customer_state	total_freight_value	avg_freight_value
1	SP	718723.07	17.37
2	RJ	305589.31	23.95
3	PR	117851.68	23.58
4	SC	89660.26	24.82
5	DF	50625.5	23.82
6	MG	270853.46	23.46
7	PA	38699.3	39.9
8	BA	100156.68	29.83
9	GO	53114.98	26.46
10	RS	135522.74	24.95

5. Analysis based on sales, freight and delivery time.

5.A. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- $\text{time_to_deliver} = \text{order_delivered_customer_date} - \text{order_purchase_timestamp}$
- $\text{diff_estimated_delivery} = \text{order_delivered_customer_date} - \text{order_estimated_delivery_date}$

```
SELECT order_id,
       DATE_DIFF(DATE(order_delivered_customer_date),DATE(order_purchase_timestamp),DAY)
as time_to_deliver,
       DATE_DIFF(DATE(order_delivered_customer_date),DATE
(order_estimated_delivery_date),DAY) as diff_estimated_delivery
FROM Target_Dataset.orders
```

Row	order_id	time_to_deliver	diff_estimated_delivery
1	1950d777989f6a877539f53795b4c3c3	30	12
2	2c45c33d2f9cb8ff8b1c86cc28c11c30	31	-29
3	65d1e226dfaeb8cdc42f665422522d14	36	-17
4	635c894d068ac37e6e03dc54eccb6189	31	-2
5	3b97562c3aee8bdedcb5c2e45a50d5e1	33	-1
6	68f47f50f04c4cb6774570cfde3a9aa7	30	-2
7	276e9ec344d3bf029ff83a161c6b3ce9	44	4
8	54e1a3c2b97fb0809da548a59f64c813	41	4
9	fd04fa4105ee8045f6a0139ca5b49f27	37	1
10	302bb8109d097a9fc6e9cefc5917d1f3	34	5

5.B. Find out the top 5 states with the highest & lowest average freight value.

WITH TOP_5 AS

```
(SELECT c.customer_state,
        ROUND(sum(oi.freight_value)/count(DISTINCT oi.order_id),2) as
```

avg_freight_value,

```
        DENSE_RANK() OVER(ORDER BY
```

```
ROUND(sum(oi.freight_value)/count(DISTINCT oi.order_id),2) DESC) as drnk
```

```
FROM Target_Dataset.order_items as oi
```

```
INNER JOIN Target_Dataset.orders AS o
```

```
ON oi.order_id = o.order_id
```

```
INNER JOIN Target_Dataset.customers as c
```

```
ON c.customer_id = o.customer_id
```

```
GROUP BY c.customer_state),
```

BOTTOM_5 AS

```
(SELECT c.customer_state,
```

```
        ROUND(sum(oi.freight_value)/count(DISTINCT oi.order_id),2) as
```

avg_freight_value,

```
        DENSE_RANK() OVER(ORDER BY ROUND(sum(oi.freight_value)/count(DISTINCT
oi.order_id),2)) as drnk
```

```
FROM Target_Dataset.order_items as oi
```

```
INNER JOIN Target_Dataset.orders AS o
```

```
ON oi.order_id = o.order_id
```

```
INNER JOIN Target_Dataset.customers as c
```

```
ON c.customer_id = o.customer_id
```

```
GROUP BY c.customer_state)
```

```
SELECT customer_state,avg_freight_value
```

```
FROM TOP_5
```

```
WHERE drnk<=5
```


UNION ALL

SELECT customer_state, avg_freight_value

FROM BOTTOM_5

WHERE drnk <= 5

ORDER BY avg_freight_value DESC;

Row	customer_state	avg_freight_value
1	RR	48.59
2	PB	48.35
3	RO	46.22
4	AC	45.52
5	PI	43.04
6	RJ	23.95
7	DF	23.82
8	PR	23.58
9	MG	23.46
10	SP	17.37

5.C. Find out the top 5 states with the highest & lowest average delivery time.

WITH RankedStates AS (

SELECT c.customer_state,

ROUND(AVG(DATE_DIFF(DATE(order_delivered_customer_date),

DATE(order_purchase_timestamp), DAY)), 2) AS avg_del_time,

DENSE_RANK() OVER(ORDER BY

ROUND(AVG(DATE_DIFF(DATE(order_delivered_customer_date), DATE(order_purchase_timestamp), DAY)), 2) DESC) AS rank_highest,

DENSE_RANK() OVER(ORDER BY

ROUND(AVG(DATE_DIFF(DATE(order_delivered_customer_date), DATE(order_purchase_timestamp), DAY)), 2)) AS rank_lowest

FROM Target_Dataset.orders AS o

INNER JOIN Target_Dataset.customers AS c

ON o.customer_id = c.customer_id

GROUP BY c.customer_state

)

SELECT

customer_state,

avg_del_time

```

FROM RankedStates
WHERE rank_highest <= 5 OR rank_lowest <= 5
ORDER BY avg_del_time DESC;

```

Row	customer_state	avg_del_time
1	RR	29.34
2	AP	27.18
3	AM	26.36
4	AL	24.5
5	PA	23.73
6	SC	14.91
7	DF	12.9
8	MG	11.95
9	PR	11.94
10	SP	8.7

5.D. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery. You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

```

SELECT
c.customer_state,ROUND(AVG(DATE_DIFF(DATE(order_estimated_delivery_date),DATE(order_delivered
_customer_date),Day)),2) AS avg_delivery_days
FROM Target_Dataset.orders AS o
INNER JOIN Target_Dataset.customers AS c
ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_delivery_days
LIMIT 5;

```

customer_state	avg_delivery_days
AL	8.71
MA	9.57
SE	10.02
ES	10.5
BA	10.79

6. Analysis based on the payments.

6.A. Find the month on month no. of orders placed using different payment types.

```
SELECT p.payment_type,EXTRACT(MONTH FROM o.order_purchase_timestamp) as  
Month,count(DISTINCT o.order_id) as no_of_orders  
FROM Target_Dataset.orders as o  
INNER JOIN Target_Dataset.payments as p  
ON o.order_id = p.order_id  
GROUP BY p.payment_type,Month  
ORDER BY Month,no_of_orders DESC
```

payment_type ▾	Month ▾	no_of_orders ▾
credit_card	1	6093
UPI	1	1715
voucher	1	337
debit_card	1	118
credit_card	2	6582
UPI	2	1723
voucher	2	288
debit_card	2	82
credit_card	3	7682
UPI	3	1942

6.B. Find the no. of orders placed on the basis of the payment installments that have been paid.

```
SELECT payment_installments AS installments,  
COUNT(order_id) AS no_of_orders  
FROM Target_Dataset.payments AS o  
WHERE payment_installments >= 1  
GROUP BY payment_installments  
ORDER BY no_of_orders DESC;
```

Row	installments ▼	no_of_orders ▼
1	1	52546
2	2	12413
3	3	10461
4	4	7098
5	10	5328
6	5	5239
7	8	4268
8	6	3920
9	7	1626
10	9	644

VALUABLE INSIGHTS

Order Trends:

- There is a visible **year-on-year increase in orders** placed, indicating growth in e-commerce adoption.
- Monthly seasonality could indicate peak periods for sales with the **highest increase in sales in the month of November**.

Customer Behaviour:

- Most orders are placed during specific times of the day, with **Afternoon** and **Night** being the busiest periods for purchases.
- Customers are distributed unevenly across states, with a few states contributing the majority of orders with **SP** having the highest order count of **41746** and **RR** having the lowest order count of **46**.
- There seems to be a significant amount of % increase (of approx. **136.98%**) in the cost of orders from 2017 to 2018 (Including only months between **January to August**).
- State with the **highest total order prices is SP** which shows that the business is thriving in terms of total sales.

Delivery Analysis:

- A significant variation is observed in the **delivery times across states**. The average delivery time and deviations from estimated delivery times highlight areas for improvement in logistics.
- **Top 5 states (SC,DF,MG,PR,SP) with the fastest deliveries** show operational efficiency, while others may require optimization.
- The **top 5 states with highest freight** values are **RR,PB,RO,AC,PI**.
- States with higher order values and freight charges indicate regions with premium customers or higher logistics costs.
- State with the **highest total freight value is SP** which shows that the state likely has a significant volume of goods being shipped, indicating robust economic activity and possibly a large consumer base.

Payment Insights:

- **Payment types** like **credit cards** or **UPI** dominate, while others may show room for growth.
- Certain months have higher orders tied to specific payment methods, indicating promotional or seasonal effects

Each customer has only **one order**, meaning there are no repeat purchases. The number of unique customers (**COUNT(DISTINCT customer_id)**) equals the total number of orders (**COUNT(order_id)**).

ACTION ITEMS FOR THE COMPANY:

- Implement **discounts** to encourage repeat purchases.
- Target marketing efforts in **low-order states** to increase brand awareness.
- Improve logistics and delivery services in states with **low order volumes**.
- **Stock inventory & increase ads** before November to **maximize revenue**.
- Analyze **monthly sales trends** to optimize promotions year-round.
- Promote the **most popular payment methods** to boost sales.
- **Expand warehouse locations** for better regional delivery coverage.
- Focus premium offerings and **exclusive deals in high-value states**.